Date: 24-4-2023

Ch 25

25.1

Given Data:

$q = 70pC = 70 \times 10^{-12} C$

$\Delta V = 20 V$.

a) $C = ?$ b) C becomes? c) V becomes?

Solution:

a) $C = \frac{q}{\Delta V} = \frac{70 \times 10^{-12}}{20}$

$$\boxed{C = 3.5 \times 10^{-12} F \text{ or } 3.5pF}$$

b) As we know $\Rightarrow \boxed{q = 200 pC.}$

capacitance is independent to (q) charge. So, capacitance remains same

$C = 3.5pF$

c) For ΔV;

$\Delta V = q/C = \frac{200pC}{3.5pF}$

$$\boxed{\Delta V = 57 V.}$$

25.4

Given Data:

$r_1 = a = 38mm = 38 \times 10^{-3} m$

$r_2 = b = 40mm = 40 \times 10^{-3} m$

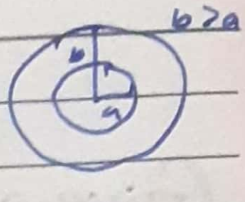
$C = ?$, $A = ?$

Date:

Solution:

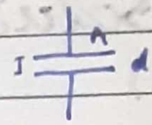
a) For C:

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}$$

(a) 

$$C = \frac{4\pi(8.85\times10^{-12})(38\times10^{-3})(40\times10^{-3})}{(40\times10^{-3})-(38\times10^{-3})}$$

$$\boxed{C = 84.5\times10^{-12}\ F \quad \text{or} \quad 84.5\,pF}$$

(b) 

b) For Area;

For d,

$$d = b-a = 40-38 = 2\,mm = 2\times10^{-3}\,m$$

Now, For Area,

$$C = \varepsilon_0 \frac{A}{d}$$

$$A = \frac{Cd}{\varepsilon_0} = \frac{(84.5\times10^{-12})(2\times10^{-3})}{(8.85\times10^{-12})}$$

$$A = 1.909\times10^{-2}\,m^2$$

$$\boxed{A = 1.91\,cm^2}$$

(25.6)

Given Data:

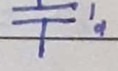
$A = 1\,m^2$

$C = 1\,F$

a) $d = ?$, b) Could the capacitor actually be constructed?

Solution:

a) For d;

(a) 

$$C = \frac{\varepsilon_0 A}{d}$$

$$d = \frac{\varepsilon_0 A}{C} = \frac{8.85\times10^{-12}(1)}{(1)}$$

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$\boxed{d = 8.85 \times 10^{-12}\, m}$

b) Constructed or not;

No, The capacitor could not be actually constructed. As the separation between the capacitor is too small to be constructed.

-(25.7)-

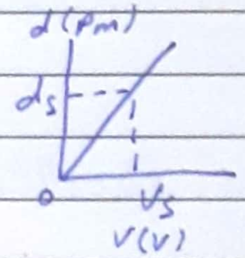
Given Data:

$n = 8.49 \times 10^{28}\, m^{-3}$

$d_s = 1\, pm = 1 \times 10^{-12}\, m$

$V_s = 20\, V.$

$C/A = ?$



Solution:

$q = Ne = (nAd)e$

As we know, $q = CV$

$CV = nAd\,e$

$\frac{C}{A} = ne\,\frac{d}{V}$

$\therefore \frac{d}{V} = \frac{1 \times 10^{-12}}{20} = 5 \times 10^{-14}$

$\frac{C}{A} = 8.49 \times 10^{28} (1.6 \times 10^{-19}) \frac{(1 \times 10^{-12})}{20}$ $\quad \therefore e = 1.6 \times 10^{-19}\, C$

$\boxed{\frac{C}{A} = 6.79 \times 10^{-4}\, Fm^{-2}}$

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{Not Included}

— (25.8) —

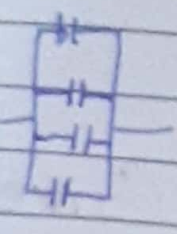
Given Data:

$C = 1\mu F = 1\times10^{-6} F$

$q = 1C$, $V = 110 V$

$n = ?$

Correct As well



Solution:

As we know,

$$C = \frac{q}{V} = \frac{1}{110}$$

$$\boxed{C = 9.09\times10^{-3} F}$$

For Parallel capacitor,

$$C = C_1 + C_2 + C_3 + \ldots$$

$$C = nC_n$$

$$n = \frac{C}{C_n}$$

Let, $C_n = 1\times10^{-6} F$.

$$n = \frac{9.09\times10^{-3}}{1\times10^{-6}}$$

$\therefore C = 9.81\times10^{-3}$

$$\boxed{n = 9.09\times10^{3}}$$

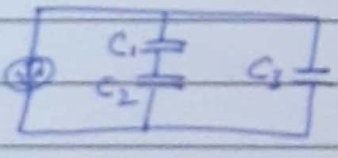
— (25.10) —

Given Data:

$C_{eq} = ?$

$C_1 = 10\mu F$, $C_2 = 5\mu F$, $C_3 = 4\mu F$

Solution:



For Capacitors in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{10} + \frac{1}{5}$$

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$$\frac{1}{C'} = \frac{1+2}{10}$$

$$\frac{1}{C'} = \frac{3}{10}$$

$$C' = \frac{10}{3}\,\mu F$$

Now, For capacitor in Parallel,

$$C_{eq} = C' + C_3 = \frac{10}{3} + 4 = \frac{10+12}{3}$$

$$\boxed{C_{eq} = \frac{22}{3}\,\mu F}$$

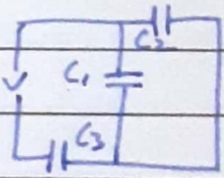
(25.11)

Given Data:

$C_1 = 10\,\mu F$, $C_2 = 5\,\mu F$, $C_3 = 4\,\mu F$

$C_{eq} = ?$

Solution:



As from given Figure,
it is clear that

$$C_{eq} = \frac{(C_1 + C_2)C_3}{C_1 + C_2 + C_3}$$

$$= \frac{(10+5)4}{10+5+4}$$

$$\boxed{C_{eq} = 3.16\,\mu F}$$

Date:

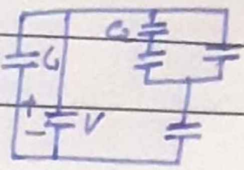
(25.14)

Given Data:

$C_{eq} = C_1 = C_2 = C_3 = C_4 = C_5 = 10\mu F$

$V = 10\,V.$

$q_1 = ?\quad , q_2 = ?$



Solution:

As we know,

a) $q_1 = C_1 V$

$= (10\times10^{-6})(10)$

$\boxed{q_1 = 1\times10^{-4}\,C}$ ✓

b) For q_2,

From given, it is clear that

$$C' = C_4 + \frac{C_2 C_3}{C_2 + C_3}$$

$$= 10 + \frac{(10)(10)}{10+10}$$

$$C' = 15\,\mu F$$

$$V' = \frac{CV}{C+C'} = \frac{10\,(20)}{10+15}$$

$$V' = 4V.$$

For V_2,

$$V_2 = \frac{V'}{2} = \frac{4}{2} = 2V$$

$\boxed{V_2 = 2V}$

Now,

$q_2 = C_2 V_2$.

$= (10\times10^{-6})(2)$

$\boxed{q_2 = 2\times10^{-5}\,C.}$ ✓

qC = q = ne $n = \frac{q}{e} = 72$ $N = \frac{q}{e}$

Date:

25.23

Given Data:

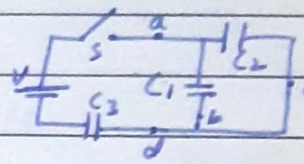
$C_1 = 4\mu F$, $C_2 = 8\mu F$, $C_3 = 12\mu F$, $V = 12V$

$n_a = ?$, $n_b = ?$, $n_c = ?$, $n_d = ?$

a) ? b) ? c) ? d) ? e) ? f) ?

Solution:

From given Figure, it is clear that



$\because || =$ ~~parallel~~ series

$$C = C_3 \,||\, (C_1 + C_2)$$
$$= 12\mu F \,||\, (4+8)$$
$$= 12\mu F \,||\, 12\mu F$$
$$\frac{1}{C} = \frac{1}{12} + \frac{1}{12}$$
$$= \frac{1+1}{12} = \frac{1}{6}$$
$$C = 6\mu F$$

For q,

$$q = CV = 6(12)$$
$$q = 72\mu C$$

In C_1,

$$q_1 = \frac{C_1}{C_1+C_2} \cdot q = \frac{4}{4+8} \cdot 72 = 24 \mu C$$

In C_2,

$$q_2 = q - q_1 = 72 - 24 = 48\mu C$$

For no of electrons,

$$1C = 6.24 \times 10^{18} \text{ electrons}$$

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a) Point a,

The series charge q through point a,

$n = 72\times10^{-6} \times 6.24\times10^{18}$

$n = 4.5\times10^{14}$ electrons.

b) Point b;

Charge q_1,

$n = (24\times10^{-6})(6.24\times10^{18})$

$n = 1.5\times10^{14}$ electrons.

c) Point c:

Charge q_2,

$n = (48\times10^{-6})(6.24\times10^{18})$

$n = 3\times10^{14}$ electrons.

d) Point d:

Same as part a,

$n = 4.5\times10^{14}$ electrons.

e) f):

As given, from point b & c, the electrons will travel upward.

Su

Date:

25.27

Given Data:

$C_1 = 1\mu F, C_2 = 2\mu F, C_3 = 3\mu F, C_4 = 4\mu F$

$V = 12V.$

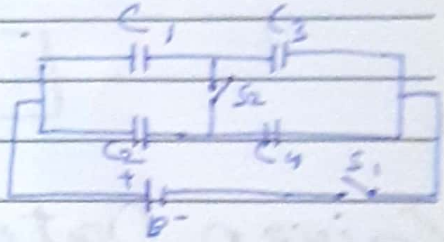
when S_1 is closed,

a) $q_1 = ?$, b) $q_2 = ?$, c) $q_3 = ?$, d) $q_4 = ?$

when both switches are closed,

e) $q_1 = ?$, f) $q_2 = ?$, g) $q_3 = ?$, h) $q_4 = ?$

Solution:



when S_1 is closed,

As from given, it is clear that, C_1 & C_3 are in series So,

$$q_1 = q_3 = \frac{C_1 C_3 V}{C_1 + C_3} = \frac{1(3)(12)}{1+3} = 9\mu C$$

$$q_2 = q_4 = \frac{C_2 C_4 V}{C_2 + C_4} = \frac{2(4)(12)}{2+4} = 16\mu C$$

a) $\boxed{q_1 = 9\mu C}$ b) $\boxed{q_2 = 16\mu C}$ c) $\boxed{q_3 = 9\mu C}$ d) $\boxed{q_4 = 16\mu C}$

when both switches are closed,

$$V_1 = \frac{C_3 + C_4}{C_1 + C_2 + C_3 + C_4} V = \frac{(3+4)12}{1+2+3+4}$$

$\boxed{V_1 = 8.4\ V}$

Now,

e) For q_1:

$$q_1 = C_1 V_1 = 1(8.4) = 8.4\mu C.$$

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$\boxed{q_1 = 8.4\,\mu C.}$

f) For q_2:

$q_2 = C_2 V_1 = 2(8.4) = 16.8\,\mu C$

$\boxed{q_2 = 16.8\,\mu C.}$

g) For q_3:

$q_3 = C_3(V - V_1) = 3(12 - 8.4) = 10.8\,\mu C$

$\boxed{q_3 = 10.8\,\mu C.}$

h) For q_4:

$q_4 = C_4(V - V_1) = 4(12 - 8.4) = 14.4\,\mu C$

$\boxed{q_4 = 14.4\,\mu C}$

25.33

Given Data:

$d = 10\,cm$

$r = \frac{d}{2} = \frac{10}{2} = 5\,cm = 5 \times 10^{-2}\,m.$

$V = 8000\,V.$

$u = ?$

Solution:

As we know,

$$u = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \left(\frac{V}{r}\right)^2$$

$$u = \frac{1}{2}(8.85 \times 10^{-12})\left(\frac{8000}{5 \times 10^{-2}}\right)^2$$

$$u = 0.11328$$

$\boxed{u = 0.11\,Jm^{-3}.}$

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Date: 14-7-18

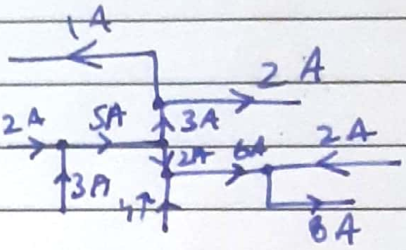
Chapter – 26

$V = \frac{E}{q}$, $i = \frac{q}{t}$

$10^{-4} - 10^{-8}$

current Travel.

Checkpoint 1:-



Ans = 8A.

Checkpoint 2:-

a)
b) ⟹ Rightward
c)

Checkpoint 4:-

$= \frac{V}{I}$

⟹ Device 1 is ohmic — where all ratio are same

⟹ Device 2 is non-ohmic → where all ratios has differrent value.

Date:

(Chapter 26)

(Q.1)

Given Data:

$I = 5\,A$, $t = 4\,min = 4 \times 60 = 240$

$q = ?$, $N = ?$

Solution:

a) columbs $\quad I = \frac{q}{t}$

$q = It = 5 \times 240$

$\boxed{q = 1200\,C}$

b) electrons $\quad N = \frac{q}{e} = \frac{1200}{1.6 \times 10^{-19}}$

$\boxed{N = 7.5 \times 10^{21}\, e}$

(Q.5)

Given Data;

$v_d = 1 \times 10^5\, ms^{-1}$, $n = 2 \times 10^8$, $q = 2e$

$J = ?$

Solution:

a) Magnitude; $J = nqv_d$

$= n\,2e\,v_d$

$= (2 \times 10^8) \cdot 2(1.6 \times 10^{-19})(1 \times 10^5)$

$\boxed{J = 6.4 \times 10^{-6}\, Am^{-2}}$

b) Direction is North.

c) Area is needed to calculate total current I in the ion beam.

Date:

-(Q.8)-

Given Data:

$i = 1.2 \times 10^{-10}$ A , $n = 8.49 \times 10^{28} m^{-3}$

$d = 2.5mm \Rightarrow r = \frac{d}{2} = \frac{2.5}{2} = 1.25mm = 1.25 \times 10^{-3} m$

$J = ?$, $V_d = ?$

Solution:

a) Current Density; $J = \frac{i}{A} = \frac{i}{\pi r^2}$ $\quad \because A = \pi r^2$

$$J = \frac{1.2 \times 10^{-10}}{\pi (1.25 \times 10^{-3})^2}$$

$$\boxed{J = 2.4 \times 10^{-5} \, Am^{-2}}$$

b) Drift Speed (V_d):

$$J = neV_d$$

$$V_d = \frac{J}{ne} = \frac{2.4 \times 10^{-5}}{(8.49 \times 10^{28})(1.6 \times 10^{-19})} = 1.76 \times 10^{-15}$$

$$\boxed{V_d = 1.8 \times 10^{-15} \, ms^{-1}}$$

-(Q.15)-

Given Data:

$N = 250$, $d = 1.3mm \Rightarrow r = \frac{d}{2} = \frac{1.3}{2} = 0.65 \times 10^{-3} m$, $r = 12 \times 10^{-2} m$

$R = ?$

Solution:

$$R = \frac{\rho L}{A} = \frac{\rho (2\pi r) N}{\pi r^2}$$

$$= \frac{1.69 \times 10^{-8} (2\pi (12 \times 10^{-2}) 250}{\pi (0.65 \times 10^{-3})^2}$$

$$\boxed{R = 2.4 \, \Omega}$$

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—(Q.17)—

Given Data:

$L = 1m$, $A = 1mm^2 = (1\times10^{-3})^2 = 1\times10^{-6}\ m^2$

$i = 4A$, $V = 2V$, $\sigma = ?$

Solution:

$$\sigma = \frac{1}{\rho} = \frac{1}{\frac{RA}{L}} = \frac{L}{RA} = \frac{L}{(\frac{V}{i})A}$$

$\because R = \rho\frac{L}{A}$

$\therefore \rho = \frac{RA}{L}$

$$\sigma = \frac{Li}{VA} = \frac{1\times4}{2(1\times10^{-6})}$$

$$\boxed{\sigma = 2\times10^{6}\ \Omega^{-1}m^{-1} \text{ or } 2\times10^{6}/\Omega m}$$

—(Q.30)—

Given Data:

$\rho = 1\Omega$, $L = 25ft$, $A = 1/4\ m^2$

$R = ?$

Solution

$$R = \frac{\rho L}{A} = \frac{1(25/1000)}{(1/4)^2}$$

$$R = \frac{25(4)^2}{1000}$$

$$\boxed{R = 0.40\ \Omega}$$

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Date:........

(Chapter 29)

(Q.1)

Given Data:

$r = 6.1\,m$, $i = 100\,A$, $B_H = 20\,\mu T$.

a) $B = ?$, b) will $= ?$

Solution:

a) For B:

$$B = \frac{\mu_0 i}{2\pi r} = \frac{(4\pi \times 10^{-7})(100)}{2\pi(6.1)}$$

$$B = 3.3 \times 10^{-6}\,T = 3.3\,\mu T$$

b) Yes, it will affect the compass reading. This is about one-sixth the magnitude of the Earth's field.

(Q.35)

Given Data:

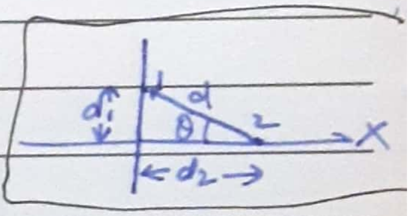
$i_1 = 4\,mA$, $i_2 = 6.80\,mA$

$d_1 = 2.4\,cm$, $d_2 = 5\,cm$,

$\left(\frac{F_B}{L}\right)_x = ?$, $d = 5.54\,cm$.

$d = \sqrt{(2.4)^2 + (5)^2}$

$d = 5.54\,cm$



Solution:

From given condition,

$$\frac{F_B}{L} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

$\therefore$ As currents are Parallel.

$$= \frac{(4\pi \times 10^{-7})(4 \times 10^{-3})(6.8 \times 10^{-3})}{2\pi(5.54 \times 10^{-2})}$$

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$$\frac{F_B}{L} = 9.8\times10^{-11}\ \frac{N}{m}$$

For x-component:

$$\left(\frac{F_B}{L}\right)x = \frac{F_B}{L}\cos\theta$$

From,

Figure, it is clear that $\theta = 30°$.

$$= (9.8\times10^{-11})\cos 30°$$ $\because \cos 30° = 0.9$

$$\boxed{\left(\frac{F_B}{L}\right)x = 8.82\times10^{-11}\,Nm^{-1}}$$

(Q, 35) (From Sol Book)

Given Data:

$i_1 = 4mA$, $i_2 = 6.80mA$

$d_1 = 2.4cm$, $d_2 = 5cm$, $r = \sqrt{d_1^2 + d_2^2} = \sqrt{d_1^2 + d_2^2}$

$\frac{F_{21}}{L}x = ?$

Solution:

$$F_{21} = \frac{\mu_0 i_1 i_2 L}{2\pi r}$$

For, x-component,

$F_{21x} = F_{21}\cos\theta$, where $\cos\theta = \frac{d_2}{\sqrt{d_1^2+d_2^2}}$

Now, Putting values,

$$F_{21x} = \frac{\mu_0 i_1 i_2 L(d_2)}{2\pi(\sqrt{d_1^2+d_2^2})(\sqrt{d_1^2+d_2^2})}$$

$$\frac{F_{21}x}{L} = \frac{\mu_0 i_1 i_2 d_2}{2\pi(d_1^2+d_2^2)}$$

$$= \frac{(4\pi\times10^{-7})(4\times10^{-3})(6.80\times10^{-3})(5\times10^{-2})}{2\pi((2.4\times10^{-2})+(5\times10^{-2})^2)}$$

$$\boxed{\frac{F_{21}x}{L} = 8.80\times10^{-11}\,Nm^{-1}}$$

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—(Q.39)—

Given Data:

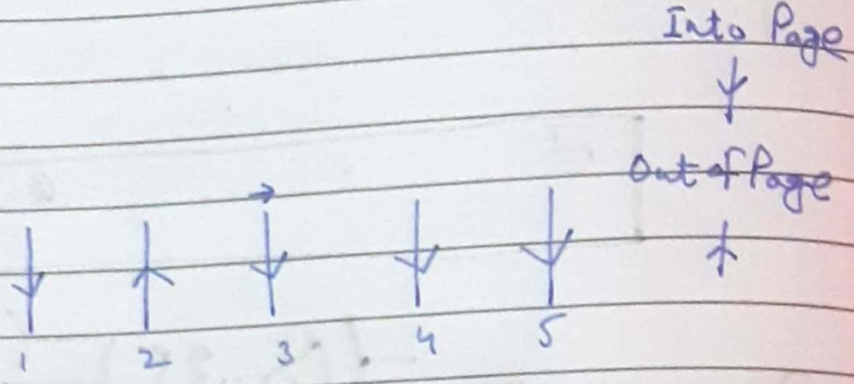
$d = 50\ cm$

$i_1 = 2\ A,\ i_2 = 4\ A,\ i_3 = 0.250\ A,\ i_4 = 4\ A,\ i_5 = 2\ A$

$\frac{|\vec{F}|}{l} = ?$

Solution:

Now,



Now,

$\overleftarrow{F_1} \quad \rightarrow F_2 + F_4 + F_5$

$F_{net} = (F_2 + F_4 + F_5 - F_1)$

From given,

$$\frac{|\vec{F}|}{l} = \frac{\mu_0 i_3}{2\pi}\left(\frac{i_2}{d} + \frac{i_4}{d} + \frac{i_5}{2d} - \frac{i_1}{2d}\right)$$

As i_5 & i_1 are in $2d$ because their distance is $2d$.

$$= \frac{(4\pi \times 10^{-7})(0.25)}{2\pi}\left(\frac{4}{50\times10^{-2}} + \frac{4}{50\times10^{-2}} + \frac{2}{2(50\times10^{-2})} - \frac{2}{2(50\times10^{-2})}\right)$$

$$= 7.96\times10^{-7}$$

$$\boxed{\frac{|\vec{F}|}{l} = 8.0\times10^{-7}\ Nm^{-1}}$$

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Date:

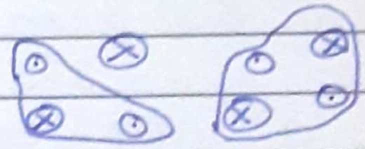
— (Q.45) —

Given Data;

$i = 2\,A$.

$\oint \vec{B}\cdot d\vec{s} = ?$

a) Path 1 = ? b) Path 2 = ?



Solution;

From given figure, it is clear that

a) Path 1:

$$\oint \vec{B}\cdot d\vec{s} = -\mu_0 i$$

$$= -(4\pi \times 10^{-7})(2)$$

$$\boxed{\oint \vec{B}\cdot d\vec{s} = -2.5 \times 10^{-6}\ T.m}$$

b) Path 2:

The net current enclosed by the path is zero (two currents are out of page and two into the page). So, we can say that,

$$\oint \vec{B}\cdot d\vec{s} = \mu_0 i_{enc}$$

$$\boxed{\oint \vec{B}\cdot d\vec{s} = 0.}$$

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Date:

— (Q.48) —

Given Data:

$R = 2.6\ cm$, $i = 8\ mA$

$d = 3R$

a) Magnitude = ? , b) Direction = ?

Solution:

a)

$$B_c = \frac{\mu_0 i_{wire}}{2\pi d} = \frac{\mu_0 i_{wire}}{2\pi(3R)}$$

$$B_c = \frac{\mu_0 i_{wire}}{6\pi R}$$

As we have, $B_{p,wire} > B_{c,wire}$.

Thus,

$$B_P = B_{p,wire} - B_{p,pipe} = \frac{\mu_0 i_{wire}}{2\pi R} - \frac{\mu_0 i}{2\pi(2R)}$$

~~[illegible]~~ setting $B_c = -B_P$. So,

$$\frac{\mu_0 i_{wire}}{6\pi R} = -\left(\frac{\mu_0 i_{wire}}{2\pi r} - \frac{\mu_0 i}{4\pi R}\right)$$

$$\frac{\mu_0 i_w}{6\pi R} = -\frac{\mu_0 i_w}{2\pi R} + \frac{\mu_0 i}{4\pi R}$$

$$\frac{\mu_0 i_w}{6\pi R} + \frac{\mu_0 i_w}{2\pi R} = \frac{\mu_0 i}{4\pi R}$$

$$\frac{\mu_0 i_w + 3\mu_0 i_w}{6\pi R} = \frac{\mu_0 i}{4\pi R}$$

$$\frac{{}_{2}\cancel{4}\,\mu_0 i_w}{{}_{3}\cancel{6}\pi R} = \frac{\mu_0 i}{4\pi R}$$

Sun

Date:

$$\frac{2\mu_0 i_w}{3\pi R} = \frac{\mu_0 i}{4\pi R}$$

$$i_w = \frac{\mu_0 i (3\pi R)}{2\mu_0 (4\pi R)}$$

$$i_w = \frac{3i}{8} = \frac{3(8\times10^{-3})}{8}$$

$$\boxed{i_w = 3\times10^{-3}\ A}$$

b) The direction is into the page.